

Francisco Castro

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Professional Summary

Results-driven scientist and technology professional and manager with extensive experience in human-centered design, data, and AI application development across industry and academic settings worldwide. Proven track record of securing \$1M+ in research funding, publishing award-winning research articles, and delivering impactful technology solutions at international organizations. Skilled at leading and collaborating in cross-functional, multi-institutional teams to build innovative technology products that reach users globally.

Experience

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|--|--------------------------------|
| New York University , Research Scientist - Human-Centered Technologies and AI | New York, NY
2022 – Present |
| <ul style="list-style-type: none">• Successfully acquired grant awards from government and private institutions for computing, technology, AI, and education research, totaling \$1M+ USD• Published and presented 40+ research articles in high-quality, peer-reviewed international conferences and journals from multi-institutional collaborative projects• Led and managed cross-functional, multi-institutional, and international project teams for research, technology development, and program implementation — projects, tools, and platforms researched/developed are at beacons.ai/franciscocastro/projects 🔗 | |
| Accenture , Senior Data and AI Advanced Application Engineering Analyst | Boston, MA
2020 - 2021 |
| <ul style="list-style-type: none">• Android Developer, Global Media and Technology – Developed and integrated an AI assistant customer experience with multi-language support using Kotlin into a customer subscription Android application• Product Owner, Multinational Consumer Electronics – Collaborated with cross-functional teams and stakeholders to manage product backlogs for the company's technology transfer into a cloud-based subscriptions management platform• Represented the company in national conferences as part of the Accenture Employee Resource Groups | |
| Amadeus , Human-Computer Interaction Research and UX Engineer Intern | Waltham, MA
2018 |
| <ul style="list-style-type: none">• Lead developer of an online airline operations response system that integrated interactive visualizations of flight information, airline delays, and situational response protocols with D3.js, Node.js, and Angular• Conducted user research to analyze the behaviors and system interactions of airline operations managers | |
| Brown University , Visiting Research Scientist | Providence, RI
2017 - 2020 |
| <ul style="list-style-type: none">• Conducted human factors research on software design systems and code design processes across multiple project sites in Boston MA, Worcester MA, and Providence RI• Published and presented 15+ research articles, including award-winning research in high-quality, peer-reviewed international conferences | |
| DataCamp , Data Science Content Development Intern | Cambridge, MA
2016 |
| <ul style="list-style-type: none">• Designed and deployed online interactive data science courses used by about 1.3 million users worldwide to date — see courses at beacons.ai/franciscocastro/projects 🔗• Developed software to automate the evaluation and feedback-delivery of user submissions | |
| Nara Institute of Science and Technology , Visiting Research Scientist | Nara, Japan
2013 |
| <ul style="list-style-type: none">• Developed the automatic speech recognition system of the Augmented Human Commu- | |

nication Laboratory by mining web and video resources speech patterns using in-house Python tools and libraries

Ateneo de Manila University, UX Designer and Researcher

Manila, Philippines

2013 - 2014

- Deployed the ehealth.ph Android mobile health information system (eHealth TABLET) to local government implementation sites across five regions in the Philippines
- Trained medical and public health professionals on mobile health system usage and health information and care-delivery digitalization
- Designed Android and web user interfaces for a traffic and air pollution analysis system and conducted usability testing to enhance the system user experience

Skills, Tools, and Technologies

Research Tools and Techniques: Experimental Design, Qualitative/Quantitative/Mixed-Methods Research, Interviews, Focus Groups, Ethnography, Card Sorting, Survey Design and Platforms (Qualtrics, SurveyMonkey), Qualitative Data Analysis Software (NVivo, ATLAS.ti, Dedoose), Research Management Platforms (Zotero, Mendeley), Google Colab

Management and Productivity Platforms: GitHub Projects, Trello, Asana, Google Workspace, Microsoft 365, Zoom, Microsoft Teams, Slack, Miro

Project Leadership: Program Management, Project Management, Product Management, Grants Development and Management, International and Cross-organizational Collaboration

Software Development and Management: Python, JavaScript, Node.js, Java, Visual Studio Code, Git, GitHub, Bitbucket, MySQL, PostgreSQL, MongoDB, DBeaver, Anaconda, HTML, CSS, Eleventy, Bootstrap

AI Tools: Claude, Gemini, Gemini Notebook, ChatGPT

Education

PhD	Worcester Polytechnic Institute , Computer Science	2020
MS	Ateneo de Manila University , Information Technology	2014
BS	University of the Philippines Los Baños , Computer Science	2010

Selected Publications

Effects of Structured LLM-Generated Feedback on Programming Assignment Performance 2026
Association for Computational Linguistics Innovative Use of NLP for Building Educational Applications
— doi.org/10.18653/v1/2026.bea-1.42

XAIxArts Manifesto: Explainable AI for the Arts 2025
ACM CHI Conference on Human Factors in Computing Systems — doi.org/10.1145/3706599.3716227

Body & Code: Distributed Cognition Exploration Into Dance & Computing Learning 2024
ACM Conference on Creativity & Cognition — doi.org/10.1145/3635636.3656206

Time-on-Task Metrics for Predicting Performance, *Best Paper Award* 2022
ACM Technical Symposium on Computer Science Education — doi.org/10.1145/3478431.3499359

- **More publications:** **Google Scholar** — scholar.google.com/citations?user=ubFX8kwAAAAJ, **ORCID** — orcid.org/0000-0002-3940-5334, **ACM Digital Library** — dl.acm.org/profile/99658729217

Professional Organizations

Association for Computing Machinery (ACM)

- Special Interest Groups on Computer-Human Interaction (SIGCHI) and Computer Science Education (SIGCSE)
- ACM Athena Award Committee Chair

Out in Science, Technology, Engineering, and Mathematics (oSTEM)